# The Magic Number Conjecture for the Rank-Two $m = 4$ Amplituhedron

A simplicial reduction of global tile relations to finite local data

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**Abstract**

For positive external data $Z \in \mathbb{R}^{n\times 6}$, the rank-two $m = 4$ amplituhedron $\mathcal{A}_{n,2,4}(Z)$ is the image of $\mathrm{Gr}_{\geq 0}(2, n)$ under $C \mapsto CZ$. We prove that every complete multiplicity-one all-$Z$ positroid tiling of $\mathcal{A}_{n,2,4}$ has

$$M(2, n-6, 2) = \frac{1}{n-3}\binom{n-3}{2}\binom{n-3}{3} = \frac{(n-3)(n-4)^2(n-5)}{12}$$

tiles, for every admissible $n \geq 6$.

The proof separates local and global issues. Let $V_n$ be the real vector space generated by all eight-dimensional positroid source cells whose amplituhedron maps are injective for every positive $Z$. Let $L_n$ be the span of the geometric codimension-one parent relations, and let $K_n$ be the kernel of the family of all-$Z$ tile-indicator maps, modulo semialgebraic sets of dimension at most seven. A finite rank-two classification proves that every local parent relation is balanced. The main global theorem is

$$K_n = L_n \qquad (n \geq 6).$$

Its proof uses signed adjacent-column amalgamation as face maps and zero insertion as degeneracy maps. These operations make $V$, $L$, $K$, and $K/L$ into simplicial real vector spaces. Normalization then reduces the infinite problem to finitely many support cores. The normalized quotient has dimensions $1, 9, 19, 12, 0, 0$ in source label numbers $6, \ldots, 11$ and vanishes thereafter by a support bound. Since $K_9 = L_9$ is certified exactly, exactness and faithfulness of simplicial normalization force $K/L = 0$ in every degree.

The proof is computer-assisted only in finite classification and exact-rank certificates; all matrix computations are over $\mathbb{Q}$ or $\mathbb{Z}$ and are accompanied by independently checkable nonzero minors. No floating-point rank calculation, exhaustive tiling enumeration, or fixed-$Z$ extrapolation is used.

**Status of this draft.** This manuscript reconstructs a computer-assisted proof that has undergone independent internal proof audits and exact certificate replays. It has not yet undergone external peer review. The finite certificates referred to below are part of the supplementary computation and should be retained with any circulated version.

## 1 Introduction

The amplituhedron $\mathcal{A}_{n,k,m}$ was introduced as a positive-geometric model for scattering amplitudes and has since developed a substantial combinatorial theory. Karp–Williams–Zhang conjectured

that for even $m$ every positroid tiling of $\mathcal{A}_{n,k,m}$ has cardinality

$$M\left(k, n-k-m, \frac{m}{2}\right),$$

the MacMahon number counting plane partitions in a $k \times (n-k-m) \times (m/2)$ box [4]. Parisi, Sherman-Bennett, Tessler, and Williams proved the conjecture for $m = 2$ using T-duality and Parke–Taylor weights [7]. For $m = 4$, Even-Zohar, Lakrec, Parisi, Tessler, Sherman-Bennett, and Williams proved that all iterated BCFW constructions give genuine tilings [5]; a companion paper exhibited the non-BCFW spurion tile [6]. Thus the cardinality problem for arbitrary positroid tilings is strictly stronger than a statement about BCFW tilings alone.

The purpose of this paper is to settle the first higher-$m$ rank slice: $k = 2$, $m = 4$, and arbitrary $n$. The main result is the following.

**Theorem 1.1** (Magic Number for $\mathcal{A}_{n,2,4}$)**.** *For every $n \geq 6$, every complete multiplicity-one all-$Z$ positroid tiling $\mathcal{T}$ of $\mathcal{A}_{n,2,4}$ has*

$$|\mathcal{T}| = M(2, n-6, 2) = \frac{1}{n-3}\binom{n-3}{2}\binom{n-3}{3} = \frac{(n-3)(n-4)^2(n-5)}{12}.$$

The proof is organized around relations among tile indicators. For each $n$ we define a free tile space $V_n$, the span $L_n$ of local relations coming from nine-dimensional positroid parents, and the full all-$Z$ indicator kernel $K_n$. The local theorem proves that every vector in $L_n$ has coefficient sum zero. The global problem is therefore reduced to proving

$$K_n = L_n.$$

At rank two, this equality admits an unexpectedly finite proof. Adjacent column amalgamation and zero insertion turn the system into a simplicial vector space. Passing to the normalized quotient discards all generators obtained by inserting loops. A classification of all rank-two $m = 4$ tiles shows that a tile has at most ten nonloop labels. Consequently no new normalized generators can occur after a finite degree. Two exact finite-rank computations at label numbers 10 and 11, together with the previously certified equality $K_9 = L_9$, close the global argument.

### Outline

Section 2 fixes conventions. Section 3 gives the finite membership-core theorem. Section 4 states the complete local parent theorem and explains why all local relations are balanced. Section 5 proves geometric descent under adjacent amalgamation. Section 6 establishes the simplicial identities. Section 7 gives the finite normalized-rank calculation, and Section 8 proves $K_n = L_n$ for all $n$. The Magic Number formula is deduced in Section 9. Appendices summarize the exact finite certificates and reproducibility requirements.

## 2 Positive Grassmannians, tiles, and indicator relations

### 2.1 Rank-two positroid cells

A point of $\mathrm{Gr}(2, n)$ is represented by a full-rank $2 \times n$ matrix $C = (c_1, \ldots, c_n)$ modulo left multiplication by $\mathrm{GL}_2$. Write $\Delta_{ij}(C)$ for the ordered $2 \times 2$ Plücker coordinate. The totally nonnegative Grassmannian is

$$\mathrm{Gr}_{\geq 0}(2, n) = \{C : \Delta_{ij}(C) \geq 0 \text{ for } i < j\}.$$

Its positroid stratification is recalled from Postnikov [1]. In rank two a cell is determined by its loops together with cyclically ordered parallel classes among the nonloop columns. If $q$ is the number of nonloop labels and $s$ the number of cyclic parallel classes, then

$$\dim S = q + s - 4. \tag{1}$$

### 2.2 The amplituhedron map

Fix a positive matrix $Z \in \mathbb{R}^{n\times 6}$, meaning that all increasing $6\times 6$ minors are positive. The rank-two $m = 4$ amplituhedron map is

$$f_Z : \mathrm{Gr}_{\geq 0}(2, n) \longrightarrow \mathrm{Gr}(2, 6), \qquad C \longmapsto CZ,$$

with $CZ$ interpreted by row span. Its image is $\mathcal{A}_{n,2,4}(Z)$, which has dimension 8.

**Definition 2.1** (All-$Z$ tile)**.** *An eight-dimensional positroid cell $S \subset \mathrm{Gr}_{\geq 0}(2, n)$ is an* all-$Z$ tile source *if $f_Z|_S$ is injective for every positive $n \times 6$ matrix $Z$. We write*

$$T_S(Z) = f_Z(\overline{S}) \subset \mathcal{A}_{n,2,4}(Z)$$

*for its closed tile image.*

The all-$Z$ convention is essential throughout. A cell that is injective for one chosen positive matrix but not uniformly in $Z$ is not included in the tile universe used here.

**Definition 2.2** (Complete multiplicity-one all-$Z$ tiling)**.** *A finite collection $\mathcal{T}$ of distinct all-$Z$ tile sources is a* complete multiplicity-one all-$Z$ positroid tiling *if the collection is independent of $Z$ and, for every positive $n \times 6$ matrix $Z$,*

$$\mathcal{A}_{n,2,4}(Z) = \bigcup_{S\in\mathcal{T}} T_S(Z), \qquad f_Z(S) \cap f_Z(S') = \varnothing \quad (S \neq S').$$

*Thus the closed tiles cover the entire amplituhedron, and their open images are pairwise disjoint. In particular, $\sum_{S\in\mathcal{T}} \mathbf{1}_{T_S(Z)} = \mathbf{1}_{\mathcal{A}_{n,2,4}(Z)}$ modulo semialgebraic sets of dimension at most seven; common tile boundaries are allowed, and no face-to-face condition is imposed.*

### 2.3 Relations modulo lower-dimensional sets

Let $\mathcal{U}_n$ denote the finite set of all all-$Z$ tile sources and put

$$V_n = \mathbb{R}^{\mathcal{U}_n} = \bigoplus_{S\in\mathcal{U}_n} \mathbb{R}e_S.$$

For a fixed positive $Z$, two indicator functions are identified when they differ only on a semialgebraic subset of dimension at most 7. Define the uniform all-$Z$ indicator kernel

$$K_n = \left\{ c \in V_n : \sum_{S\in\mathcal{U}_n} c_S \mathbf{1}_{T_S(Z)} = 0 \pmod{\dim \leq 7} \text{ for every positive } Z \right\}. \tag{2}$$

The coefficients $c_S$ are independent of $Y$ and $Z$.

A complete multiplicity-one all-$Z$ tiling $\mathcal{T}$ defines a binary vector

$$\chi_{\mathcal{T}} = \sum_{S\in\mathcal{T}} e_S \in V_n.$$

If $\mathcal{T}$ and $\mathcal{T}'$ tile the same amplituhedron, then

$$\chi_{\mathcal{T}} - \chi_{\mathcal{T}'} \in K_n. \tag{3}$$

Thus controlling $K_n$ controls all tiling differences without requiring flip connectivity or enumeration of all tilings.

## 3 The complete rank-two membership classification

The global argument will use the fact that all tile sources have bounded nonloop support. We therefore record the complete rank-two classification.

**Theorem 3.1** (Membership-core theorem)**.** *Let $S \subset \mathrm{Gr}_{\geq 0}(2,n)$ be an eight-dimensional positroid cell. Let $q$ be its number of nonloop labels and let $b_1, \dots, b_s$ be the cyclic parallel-class sizes. Then $S$ is an all-$Z$ tile source if and only if either*

$$\max_i b_i = q - 5, \tag{4}$$

*or*

$$(q; s; b_1, b_2, b_3) = (9; 3; 3, 3, 3).$$

*Up to signed dihedral symmetry and insertion of loops, there are exactly eleven admissible loopless core types: ten satisfying* (4) *and the* $(3,3,3)$ *spurion.*

*Proof.* By (1), dimension eight gives $q + s = 12$, hence

$$(q, s) = (6,6), (7,5), (8,4), (9,3), (10,2).$$

If a parallel class $D$ has at most four complementary nonloop labels, then a nonzero source row functional annihilating the direction of $D$ produces a nonzero image vector supported on at most four fixed external rows. Therefore the image plane meets a fixed external $t$-space with $t \leq 4$. The corresponding incidence locus in $\mathrm{Gr}(2,6)$ has dimension at most $t+3 \leq 7$, so the eight-dimensional cell cannot map injectively to an open eight-dimensional image.

If no class has exactly five complementary labels and no class has at most four complementary labels, the five possible pairs $(q,s)$ force only two remaining profiles: $(2,2,2,2)$ and $(3,3,3)$. The former is the four-mass cell and is not all-$Z$ injective; see Galashin–Lam [3] and the exact two-preimage certificate in the supplementary computation. The latter is the spurion, whose all-$Z$ injectivity was proved in [6].

It remains to prove injectivity for the complement-five family. Delete loops and cyclically cut so that a distinguished class $D$ is last and its five complementary labels occur first. If $A_1, \dots, A_\ell$ are the successive complementary classes and $r = |D|$, then $\ell + r = 6$ and every open source point admits the chart

$$C = \begin{pmatrix} a_1 & \cdots & a_5 & 0 & \cdots & 0 \\ t_{j(1)}a_1 & \cdots & t_{j(5)}a_5 & d_1 & \cdots & d_r \end{pmatrix}, \qquad a_i, d_h > 0, \quad t_1 < \cdots < t_\ell.$$

Let $H = \mathrm{span}(Z_1, \dots, Z_5)$ and $h(X) = \det(Z_1, \dots, Z_5, X)$. The first image row lies in $H$, while the second has positive $h$-value, so $Y \cap H$ is a unique line. Its positive ray uniquely determines $a_1, \dots, a_5$ up to common scale. Put

$$V_j = \sum_{i \in A_j} a_i Z_i.$$

The six rows

$$(V_1, \dots, V_\ell, Z_{D_1}, \dots, Z_{D_r})$$

are independent because their determinant is a positive sum of increasing $6 \times 6$ minors of $Z$. Expanding any second image row of positive $h$-value in this basis recovers all slopes $t_j$ and all $d_h$ uniquely. Thus the source row space is uniquely recovered. This proves all-$Z$ injectivity for all ten complement-five cores.

The finite composition list is then obtained from the five possibilities $q + s = 12$ up to cyclic rotation and reversal. The eleven admissible core types are displayed in Table 1. Loop insertion and signed dihedral transport preserve the entire source/image problem, so the classification holds for arbitrary $n$. □

| $q$ | number of loopless cells | admissible class-size cores |
|---|---|---|
| 6 | 1 | $(1,1,1,1,1,1)$ |
| 7 | 14 | $(1,1,1,2,2)$, $(1,1,2,1,2)$ |
| 8 | 36 | $(1,1,3,3)$, $(1,2,2,3)$, $(1,2,3,2)$, $(1,3,1,3)$ |
| 9 | 30 | $(1,4,4)$, $(2,3,4)$, $(3,3,3)$ |
| 10 | 5 | $(5,5)$ |

Table 1: The eleven admissible loopless membership cores. The $(3,3,3)$ profile is the spurion; the other ten are complement-five.

**Corollary 3.2** (Support bound and tile-source count)**.** *Every all-Z tile source has at most ten nonloop labels. Moreover*

$$|\mathcal{U}_n| = \binom{n}{6} + 14\binom{n}{7} + 36\binom{n}{8} + 30\binom{n}{9} + 5\binom{n}{10}. \tag{5}$$

# 4 The local parent relation theorem

Let $P$ be a nine-dimensional positroid cell and let

$$\mathcal{F}(P) = \{S : \dim S = 8,\ S \lessdot P,\ S \in \mathcal{U}_n\}$$

be its admissible codimension-one source facets. Define

$$\mathcal{R}_{\text{all}}(P) = \left\{ c \in \mathbb{R}^{\mathcal{F}(P)} : \sum_{S \in \mathcal{F}(P)} c_S \mathbf{1}_{T_S(Z)} = 0 \pmod{\dim \le 7} \text{ for every positive } Z \right\}.$$

Let $L_n \subset V_n$ be the span of all such geometric one-parent relations, together with their loop and signed-dihedral transports.

**Theorem 4.1** (Complete one-parent theorem)**.** *For every nine-dimensional rank-two positroid parent $P$ and every $n \ge 7$, the space $\mathcal{R}_{\text{all}}(P)$ is either zero or the one-dimensional span of a primitive balanced vector. Up to loop deletion and dihedral relabeling, the 29 loopless parent cores split as follows:*

| *sector* | *number of cores* | *admissible facets* | *relation space* |
|---|---|---|---|
| *collapsed parent image* | *12* | *0* | 0 |
| *complement-five circuit* | *10* | *6* | $\mathbb{R}\varepsilon$, $3 \leftrightarrow 3$ |
| *reflection/spurion* | *5* | *10* | $\mathbb{R}\varepsilon$, $5 \leftrightarrow 5$ |
| *hostile four-mass* | *2* | *8* | 0 |

*In particular, every $r \in L_n$ satisfies*

$$\sum_{S \in \mathcal{U}_n} r_S = 0. \tag{6}$$

*Proof structure.* For rank two, a nine-dimensional parent has $q + s = 13$. Up to cyclic rotation and reversal this yields 29 loopless core types. The source facets of every core are enumerated exactly and are classified using Theorem 3.1: each facet is either an all-$Z$ tile, collapsed by the class-complement obstruction, or the noninjective four-mass type.

For ten cores a distinguished class has exactly five complementary labels. The positive fiber is an affine interval whose six actual wall functions have alternating derivatives, obtained from the Gale vector of a positive $7 \times 6$ matrix. This gives a multiplicity-one $3 \leftrightarrow 3$ relation. Four further cores admit an orientation-reversing source reflection paired with an orientation-preserving target action; the ten facets therefore pair with opposite image signs, giving $5 \leftrightarrow 5$. The remaining balanced core is the repaired spurion pattern, whose wall-sign sum is zero by an ordered-minor/Plücker calculation.

It remains to show that no additional unbalanced relation exists on the same facet set. For the six-wall cores, opposite-side open overlaps form $K_{3,3}$, forcing every coefficient relation to be a scalar multiple of the known balanced vector. For each ten-wall core, nine exact rational open-overlap witnesses form a spanning tree on the ten facets; the nine edge equations have rank nine and therefore one-dimensional kernel. For the two hostile cores, eight exact target neighborhoods produce an $8 \times 8$ tile-indicator matrix of determinant 1, so their relation spaces are zero. All determinants and membership/nonmembership decisions are exact and independently replayed. Loop insertion and signed dihedral transport extend the finite core statement to all $n$. □

**Remark 4.2** (Computer-assisted scope)**.** *The finite core enumeration, the ten-wall spanning-tree certificates, and the two hostile determinant certificates are exact finite computations. No floating-point geometry is used. The symbolic complement-five family and the transport arguments are uniform in positive $Z$.*

## 5 Adjacent amalgamation and geometric descent

We now introduce the global structure that makes the arbitrary-$n$ problem finite. For a source matrix $C = (c_1, \dots, c_n)$ define ordinary adjacent amalgamation by replacing two adjacent columns by their sum. The cyclic wrap requires the rank-two sign twist.

For $0 \le i < n - 1$, set

$$d_i(c_1, \dots, c_n) = (c_1, \dots, c_{i+1} + c_{i+2}, \dots, c_n), \tag{7}$$

and set

$$d_{n-1}(c_1, \dots, c_n) = (c_1 - c_n, c_2, \dots, c_{n-1}). \tag{8}$$

On the free tile space, retain the resulting source label only when it is again an admissible eight-dimensional tile source; otherwise define the image to be zero.

**Theorem 5.1** (Geometric amalgamation)**.** *For every $n \ge 7$, the maps $d_i : V_n \to V_{n-1}$ have the following properties.*

*(i) Each source face has a single exact positroid-cell image. If source dimension is preserved and the output is admissible, the map between the open source cells is a positive rational isomorphism.*

*(ii) If an eight-dimensional output is discarded, then it is collapsed, not four-mass; its target image has dimension at most seven for every positive downstairs external matrix.*

*(iii) The maps descend to all-Z indicator kernels:*

$$d_i(K_n) \subseteq K_{n-1}.$$

*(iv) They descend to the local relation spans:*

$$d_i(L_n) \subseteq L_{n-1}.$$

*Proof.* We give the ordinary last-pair case; all other ordinary positions follow by signed cyclic transport, and the wrap is the same geometric operation in twisted cyclic coordinates.

Write the last two source columns as $a, b$ and their amalgamation as $w = a + b$. Every output ordered minor is either an unchanged source minor or

$$\Delta_{i,w} = \Delta_{i,n-1} + \Delta_{i,n} \qquad (i < n-1).$$

Since all summands are nonnegative, there is no cancellation; the full source zero pattern therefore determines a unique output positroid label. Rank loss on the nonnegative Grassmannian can occur only at the coordinate two-plane supported on the merged pair. Indeed, if every minor involving $w$ vanishes, the nonnegative summands vanish separately, and any surviving outside column would force all columns to be parallel, contradicting rank two.

A direct case analysis on loops, same-class amalgamation, and distinct parallel classes shows that every dimension-preserving nonloop face admits a rational inverse. Geometrically, the output direction $w$ lies strictly between the two surviving old directions; its positive decomposition in those two rays uniquely recovers the original pair up to source row-space gauge. Faces involving a disappearing singleton or same-class pair lower dimension and are treated only as lower-dimensional outputs.

By Theorem 3.1, a dimension-preserving nonloop face creates a singleton class, so an inadmissible eight-dimensional output cannot be the four-mass profile $(2, 2, 2, 2)$. Every other excluded eight-cell has a class with at most four complementary nonloops and is therefore collapsed by the incidence-dimension argument used in Theorem 3.1.

Fix an arbitrary positive downstairs external matrix $W$. It can be realized as a positive degeneration of upstairs external matrices $Z(\epsilon)$ in which the relevant adjacent rows coalesce as $\epsilon \to 0^+$. One may choose the perturbation so that every increasing six-minor remains positive for sufficiently small $\epsilon$.

The only source rank-loss limit is the coordinate plane supported on the merged pair. Every limiting target from this locus contains the duplicated external row and therefore lies in a Schubert locus of dimension four. All collapsed or lower-dimensional output cells contribute target sets of dimension at most seven. Away from the finite union of these exceptional sets and source-boundary/critical images, surviving tile indicators are locally constant under the degeneration: membership follows from the rational source isomorphism and the parameter-dependent inverse function theorem, while closed nonmembership follows by compactness of source closures.

Now let $r \in K_n$. On a generic downstairs open neighborhood, every upstairs tile indicator appearing in $r$ stabilizes to the corresponding downstairs indicator, with multiplicity retained. If $d_i(r)$ were nonzero on that neighborhood, the same nonzero value would persist for all small positive $\epsilon$, contradicting the all-$Z$ relation defining $r$. Hence $d_i(K_n) \subseteq K_{n-1}$.

Finally take a primitive one-parent relation. The exact image of its nine-dimensional parent is either nine-dimensional, eight-dimensional, or smaller. In the first case the surviving facet relation belongs to the already classified one-parent kernel of the downstairs parent. In the second case all surviving facets have the same eight-dimensional output label, and the resulting scalar relation must be zero on its nonempty open image. In the last case every surviving output is lower-dimensional. Thus $d_i(L_n) \subseteq L_{n-1}$ without using the global equality to be proved. □

## 6 The twisted simplicial structure

For $0 \le i < n$, let $s_i : V_n \to V_{n+1}$ insert a zero source column after $c_{i+1}$. This is loop insertion. On source closures,

$$(j_I C)Z = CZ_I, \tag{9}$$

so zero insertion preserves all-$Z$ tile relations and local parent relations exactly.

**Theorem 6.1** (Simplicial structure)**.** *Set $V_n = L_n = K_n = 0$ for $n \le 5$. With simplicial degree $p = n - 1$, the spaces*

$$V_{p+1}, \qquad L_{p+1}, \qquad K_{p+1}, \qquad \mathcal{Q}_p := V_{p+1}/L_{p+1}, \qquad \mathcal{H}_p := K_{p+1}/L_{p+1}$$

*form simplicial real vector spaces, and $\mathcal{H}$ is a simplicial subobject of $\mathcal{Q}$.*

*Proof.* At the raw column level, the face and degeneracy maps satisfy

$$d_i d_j = d_{j-1} d_i \qquad (i < j), \tag{10}$$

$$d_i s_j = \begin{cases} s_{j-1} d_i, & i < j, \\ \mathrm{id}, & i = j \text{ or } i = j+1, \\ s_j d_{i-1}, & i > j+1, \end{cases} \tag{11}$$

$$s_i s_j = s_{j+1} s_i \qquad (i \le j). \tag{12}$$

For non-wrap faces these are associativity and insertion identities. The two endpoint computations involving the wrap are

$$(x_0 - x_{n-1}) + x_1 = (x_0 + x_1) - x_{n-1},$$

and

$$(x_0 - x_{n-1}) - x_{n-2} = x_0 - (x_{n-2} + x_{n-1}).$$

The face–degeneracy identities at the cyclic boundary are checked in the same signed convention.

It remains to justify truncation: a raw intermediate that has been sent to zero must never become an admissible eight-cell later on one side of an identity. Rank loss and source-dimension drop cannot be repaired by later faces or zero insertion. If the discarded intermediate is an inadmissible eight-cell, Theorem 3.1 and the singleton created by a dimension-preserving nonloop amalgamation imply that it is collapsed, not four-mass. Its class-complement obstruction persists under every subsequent dimension-preserving face: amalgamating away from the obstructing class decreases the complement, while amalgamating the obstructing class decreases both its size and total support. Hence a discarded intermediate remains discarded. The raw equalities therefore descend to the truncated free tile spaces.

By Theorem 5.1, faces preserve $K$ and $L$. By (9), degeneracies preserve $K$ and $L$. The simplicial identities consequently restrict to these subspaces and descend to their quotients. □

## 7 Normalization and the finite quotient

For a simplicial real vector space $X$, let $N(X)$ denote its normalized chain complex. We use the standard facts that $N(X)_p$ is canonically isomorphic to the quotient of $X_p$ by the span of degeneracies from degree $p - 1$, that normalization is exact, and that it detects the zero simplicial object. These are standard consequences of Dold–Kan; see the Stacks Project [8, 9, 10].

Since $s_i$ inserts a loop in one of positions $2, \dots, n$, normalization of the quotient $Q_n := V_n/L_n$ gives

$$N(\mathcal{Q})_{n-1} \cong V_n \Big/ \left(L_n + \operatorname{span}\{e_S : S \text{ has a loop in } \{2, \dots, n\}\}\right). \tag{13}$$

A tile whose only loop is at position 1 survives this quotient.

The support bound in Corollary 3.2 now makes the normalized problem finite. The surviving columns at label number $n$ are exactly the loopless $n$-label tiles together with the $(n-1)$-label loopless tiles inserted with their unique loop at position 1. The only projected parent rows that can be nonzero come from loopless parents or parents with their unique loop at position 1.

**Proposition 7.1** (Exact normalized ranks)**.** *The normalized quotients have the following exact finite presentations:*

| $n$ | *surviving tile columns* | *projected parent rows* | *row rank* | $\dim N(\mathcal{Q})_{n-1}$ |
|---|---|---|---|---|
| *6* | *1* | *0* | *0* | *1* |
| *7* | *15* | *7* | *6* | *9* |
| *8* | *50* | *47* | *31* | *19* |
| *9* | *66* | *121* | *54* | *12* |
| *10* | *35* | *141* | *35* | *0* |
| *11* | *5* | *46* | *5* | *0* |

*The displayed maximal nonzero minors can be chosen with determinants*

$$1, -1, 1, -1, -1, 1$$

*in order. For $n \geq 12$ there are no normalized tile generators. Consequently*

$$N(\mathcal{Q})_p = 0 \qquad (p \geq 9). \tag{14}$$

*Proof.* The column and row sets are determined exhaustively from the eleven membership cores and the 29 parent cores. Rows from parents carrying a loop in positions $2, \dots, n$ vanish after projection because every facet retains that loop. The remaining rows are literal transports of the finite parent prototypes.

All ranks are computed over $\mathbb{Q}$. For $n = 10$ an independently reconstructed $35 \times 35$ submatrix has determinant $-1$, and for $n = 11$ an independently reconstructed $5 \times 5$ submatrix has determinant 1. Rational row reduction gives the corresponding upper bounds, hence these matrices have full column rank. The lower-$n$ ranks are checked by the same exact procedure. Finally, every tile has at most ten nonloop labels by Corollary 3.2; for $n \geq 12$, every tile therefore has a loop in positions $2, \dots, n$ and dies in the normalized quotient. □

**Remark 7.2.** *Only the two high-degree zero statements at $n = 10, 11$ and the support cutoff are needed for the all-$n$ vanishing argument. The lower normalized dimensions are useful for the rank formula below but are not an extrapolation hypothesis.*

# 8 Uniform global generation

The finite base case at $n = 9$ is the last ingredient. It is obtained by a complete exact indicator-kernel certificate using the 942 all-$Z$ tile sources at $n = 9$. The local relation matrix has rank 470. An exact $472 \times 942$ open-neighborhood indicator matrix has rank 472, with a $472 \times 472$ minor of

determinant 4, and annihilates every local relation. A separate implementation checks all 444,624 membership decisions, including the spurion inverse. Therefore

$$K_9 = L_9. \tag{15}$$

We isolate the resulting infinite statement.

**Theorem 8.1** (Uniform indicator generation)**.** *For every $n \geq 6$,*

$$K_n = L_n.$$

*Proof.* Let

$$\mathcal{H}_p = K_{p+1}/L_{p+1}, \qquad \mathcal{Q}_p = V_{p+1}/L_{p+1}.$$

By Theorem 6.1, $\mathcal{H} \hookrightarrow \mathcal{Q}$ is a monomorphism of simplicial real vector spaces.

First, $\mathcal{H}_p = 0$ for $p \leq 8$. Indeed, (15) gives the claim at label number 9. For every smaller label number, insert zero columns up to nine labels and then apply the corresponding face maps. The face–degeneracy identities give left inverses to the insertion, while Theorem 5.1 guarantees that the faces preserve $L$. Hence any nonzero class in $K_r/L_r$ for $r < 9$ would produce a nonzero class in $K_9/L_9$, a contradiction.

Apply the exact normalization functor. Since $\mathcal{H} \hookrightarrow \mathcal{Q}$, exactness gives

$$N(\mathcal{H}) \hookrightarrow N(\mathcal{Q}).$$

For $p \leq 8$ the source $\mathcal{H}_p$ is already zero, so $N(\mathcal{H})_p = 0$. For $p \geq 9$, (14) gives $N(\mathcal{Q})_p = 0$, and hence $N(\mathcal{H})_p = 0$ as well. Thus

$$N(\mathcal{H}) = 0$$

in every degree. Faithfulness of normalization (equivalently, Dold–Kan) forces $\mathcal{H} = 0$. Therefore $K_n/L_n = 0$ for every $n \geq 6$. □

### 8.1 An all-$n$ quotient-rank formula

The normalized decomposition of a simplicial vector space is indexed by order-preserving (weakly increasing) surjections $[n-1] \twoheadrightarrow [p]$, where $[p] = \{0, \ldots, p\}$. There are $\binom{n-1}{p}$ such surjections. Using Proposition 7.1, one obtains

**Corollary 8.2.** *For every $n$,*

$$\dim(V_n/L_n) = \binom{n-1}{5} + 9\binom{n-1}{6} + 19\binom{n-1}{7} + 12\binom{n-1}{8}, \tag{16}$$

*where binomial coefficients outside their natural range are zero. Consequently*

$$\operatorname{rank} L_n = \dim K_n = |\mathcal{U}_n| - \dim(V_n/L_n),$$

*with $|\mathcal{U}_n|$ given by* (5)*.*

## 9 The Magic Number corollary

*Proof of Theorem 1.1.* Let $\mathcal{T}$ and $\mathcal{T}'$ be complete multiplicity-one all-$Z$ tilings. By (3),

$$\chi_{\mathcal{T}} - \chi_{\mathcal{T}'} \in K_n.$$

By Theorem 8.1, this difference lies in $L_n$. The augmentation functional

$$\epsilon : V_n \to \mathbb{R}, \qquad \epsilon\left(\sum_S c_S e_S\right) = \sum_S c_S$$

vanishes on $L_n$ by (6). Therefore

$$|\mathcal{T}| - |\mathcal{T}'| = 0.$$

All complete all-$Z$ tilings thus have the same number of tiles.

A standard BCFW tiling exists for every admissible $n$ and has the Narayana cardinality

$$\frac{1}{n-3}\binom{n-3}{2}\binom{n-3}{3}$$

for $k = 2$, by the $m = 4$ BCFW tiling theorem and the established BCFW count [4, 5]. Comparing an arbitrary tiling to a standard BCFW tiling gives the stated formula. The algebraic simplification to $(n-3)(n-4)^2(n-5)/12$ is immediate. □

**Remark 9.1** (Scope)**.** *The theorem concerns constant-coefficient all-Z indicator relations and complete all-Z positroid tilings. It does not assert that the full fixed-Z indicator kernel equals $L_n$ for every individual positive $Z$, that the flip graph is connected, or that $L_n$ is spanned by actual global tiling differences for arbitrary $n$. It also makes no claim for $k > 2$.*

## 10 Discussion

The proof separates into a rank-two local classification and a global simplicial argument. The local step is strongly rank-specific: it uses the description of $\mathrm{Gr}_{\geq 0}(2, n)$ by cyclic parallel classes and a finite classification of nine-dimensional parents. By contrast, the final mechanism—adjacent amalgamation, loop insertion, normalization, and support-bounded vanishing—is formulated at the level of source-cell relations and may admit higher-rank analogues. Determining which part of this structure survives for $k \geq 3$ is a natural next problem toward the full $m = 4$ Magic Number Conjecture.

The proof also contrasts with the $m = 2$ argument of [7]. There, a Parke–Taylor weight gives a direct unit valuation on tiles. Here the augmentation functional becomes well-defined only after the global relation theorem $K_n = L_n$. The finite $n = 8, 9$ indicator certificates do admit signed rational unit functionals, but their specific weights do not appear to depend only on simple tile profile data. A canonical $m = 4$ weight formula, if one exists, would provide an alternative route to higher rank.

## A Exact finite certificates

The proof uses three finite families of exact certificates.

### A.1 Local parent certificates

The 29 loopless nine-dimensional parent cores are exhaustively obtained from $q + s = 13$. For the five ten-wall balanced cores, exact open-overlap witnesses yield rank-nine spanning-tree constraint matrices. For the two hostile cores, exact $8 \times 8$ indicator matrices have determinant 1. All source coordinates, positive minors, image equalities, and nonmembership checks are rational.

### A.2 The $n = 9$ kernel certificate

At $n = 9$ the complete tile universe has 942 columns. The transported local relation matrix has 693 rows and exact rank 470. The open-region indicator matrix has 472 independent rows; a selected $472 \times 472$ minor has determinant 4. Since the indicator rows annihilate all local relations,

$$L_9 \subseteq K_9 \subseteq \ker A_9,$$

and both $L_9$ and $\ker A_9$ have dimension 470. Thus $K_9 = L_9$. The independent replay checks $472 \cdot 942 = 444{,}624$ membership decisions.

### A.3 Normalized $n = 10, 11$ certificates

After quotienting by degeneracies, label number 10 has 35 surviving tile columns and 141 projected parent rows. A literal-basis independent reconstruction gives a $35 \times 35$ minor of determinant $-1$. At label number 11, the corresponding $5 \times 5$ minor has determinant 1. These certify the vanishing of the normalized quotient in the two critical highest degrees before the support cutoff.

## B Reproducibility and audit trail

The supplementary proof material is organized by theorem dependency:

- `research/run4/LOCAL_PARENT_THEOREM.md`: complete local parent theorem and its finite certificates;
- `research/run6/MEMBERSHIP_CORES.md` and `research/run6/SPURION_INVERSION.md`: complete tile membership classification and the two inverse mechanisms;
- `research/run6/N9_INDICATOR_KERNEL.md`: exact $n = 9$ base kernel theorem;
- `research/run7/AMALGAMATION_PROOF.md`: geometric face maps and descent of $K$ and $L$;
- `research/run7/SIMPLICIAL_AMALGAMATION.md`: signed simplicial identities and truncation proof;
- `research/run7/GLOBAL_GENERATION.md`: global normalization argument;
- `results/run7/normalized_core_ranks.json` and `results/run7/normalized_independent_review.json`: exact finite normalized matrices and independent determinant replay.

All promoted rank claims are over $\mathbb{Q}$ or $\mathbb{Z}$. The final campaign validation preserves all prior artifacts byte-for-byte and runs the complete regression suite.

**Supplementary repository: [URL TO BE ADDED].** The accompanying archive is `magic-number-proof-material.zip`.

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This draft was developed with extensive AI-assisted proof search, formalization of finite cases, exact computation, and adversarial proof review. The author is responsible for the mathematical claims. Independent human verification is welcome; this manuscript has not yet undergone external peer review.

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